Agentic AI for Phase-Field Simulation and Multitask Materials Science

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Section 1

AutoMOOSE: An Agentic AI for Autonomous Phase-Field Simulation

Basic Information

- **Title**: *AutoMOOSE: An Agentic AI for Autonomous Phase-Field Simulation.*
- **Code**: `https://github.com/sukritimanna/AutoMOOSE`.
- **Publication time**: First posted on arXiv on 22 March 2026; updated on 7 August 2026.
- **Journal/Venue**: Not yet published in a peer-reviewed journal.
- **Authors**: Sukriti Manna[1,2,*], Henry Chan[2], Subramanian K. R. S. Sankaranarayanan[1,2,*]
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Difficulties in Phase-Field Simulation Automation

- Complex multiphysics inputs with strict dependencies; errors cause solver failure or silent physical invalidity.
- Execution success $\neq$ physical correctness; requires conservation-law and scaling verification.
- Reproducibility limited by undocumented parameters, stochasticity, finite-size effects, and HPC scaling.

Prior Work: MooseAgent

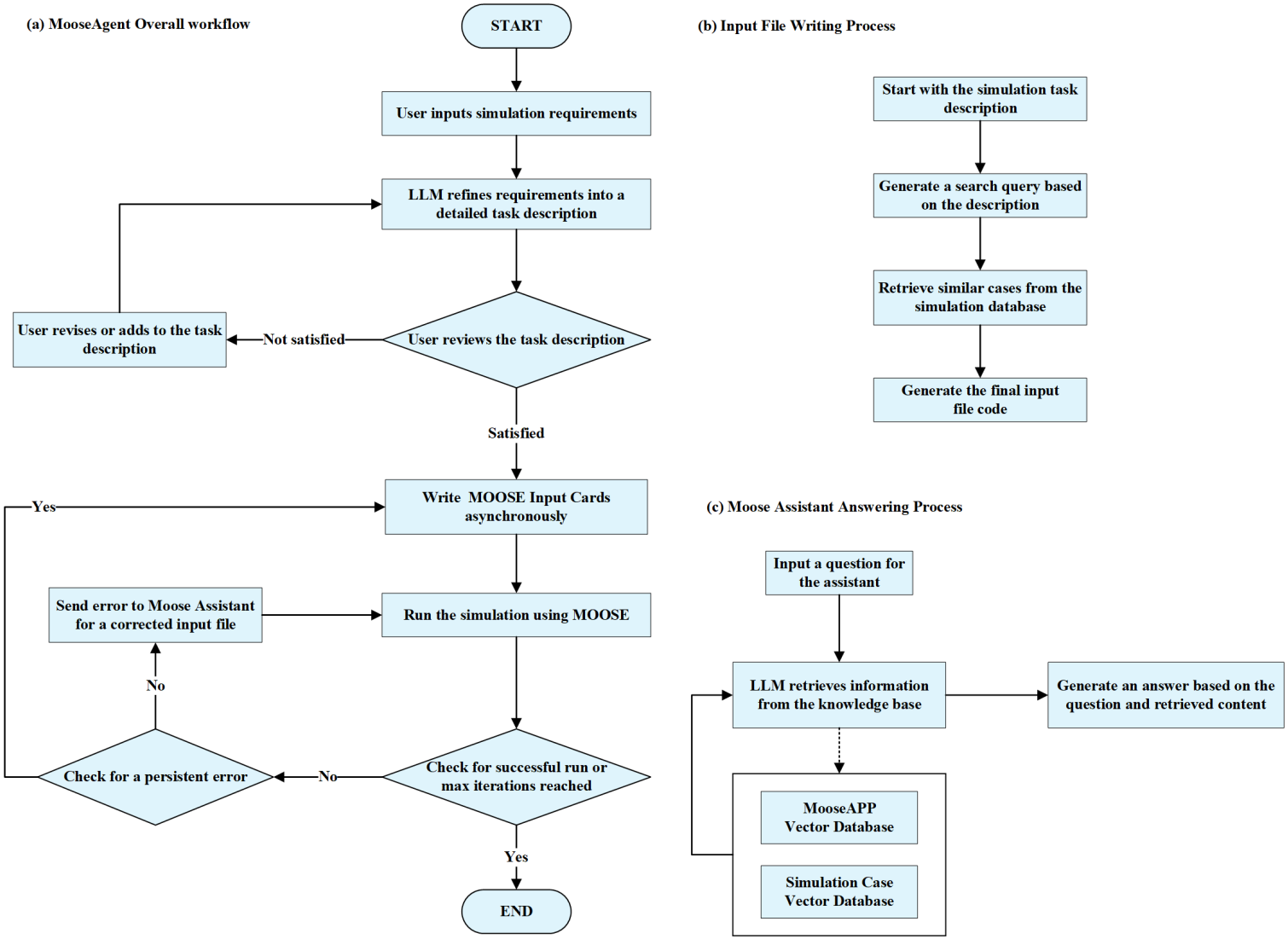


Figure 1: Framework of MooseAgent [Zhang et al., 2025].

Limitations

- Validates execution success only, not physical correctness.
- Failure recovery relies on unconstrained LLM retries (up to 3 iterations), often stuck in loops.
- Evaluation is not pre-registered or machine-checkable, with no independent physics verification.

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Mathematical Framework

- Utilizes the multiphase Allen–Cahn model [Moelans et al., 2008] as the representative physics.
- Represent each grain i by an order parameter η_i $(i = 1, \ldots, N_{\text{op}})$, equals 1 inside the grain and 0 o.w. Each η_i evolves by the Allen–Cahn equation,

$$\frac{\partial \eta_i}{\partial t} = -L(T)\frac{\delta F}{\delta \eta_i}, \tag{1}$$

where $L(T)$ is the temperature-dependent kinetic coefficient and F the total free energy,

$$F = \int_V \left[f_{\text{loc}}(\boldsymbol{\eta}) + \frac{\kappa}{2} \sum_{i=1}^{N_{\text{op}}} |\nabla \eta_i|^2 \right] dV, \tag{2}$$

with the local free energy density in the Moelans form,

$$f_{\text{loc}} = \mu \left(\sum_i \left(\frac{\eta_i^4}{4} - \frac{\eta_i^2}{2} \right) + \gamma \sum_i \sum_{j>i} \eta_i^2 \eta_j^2 + \frac{1}{4} \right), \tag{3}$$

where μ is the free energy weight and $\gamma = 1.5$.

Mathematical Framework (cont'd)

- The model coefficients $\{L, \mu, \kappa\}$ depend on the user specified parameters — grain boundary energy σ, interface width w_{GB}, pre-exponential mobility M_0, activation energy Q, temperature T, — through

$$L = \frac{4M_{\mathrm{GB}}}{3w_{\mathrm{GB}}}, \qquad \mu = \frac{6\sigma}{w_{\mathrm{GB}}}, \qquad \kappa = \frac{3}{4}\sigma w_{\mathrm{GB}}. \tag{4}$$

  where

$$M_{\mathrm{GB}}(T) = M_0 \exp\left(-\frac{Q}{k_{\mathrm{B}}T}\right) \tag{5}$$

- The grain count $N(t)$ obeys the grain coarsening law (in two dimensions):

$$\frac{1}{N(t)} - \frac{1}{N_0} = \tilde{k}(T)t, \tag{6}$$

  where $N_0 = N(0)$. $\tilde{k}(T)$ is obtained from the slope of $N^{-1}(t)$ against t, which follows,

$$\tilde{k}(T) = \tilde{k}_0 \exp\left(-\frac{Q}{k_{\mathrm{B}}T}\right), \tag{7}$$

- The activation energy Q recovered from this macroscopic fit must match the user-specified input, serving as a closed-loop consistency check.

Architecture Overview

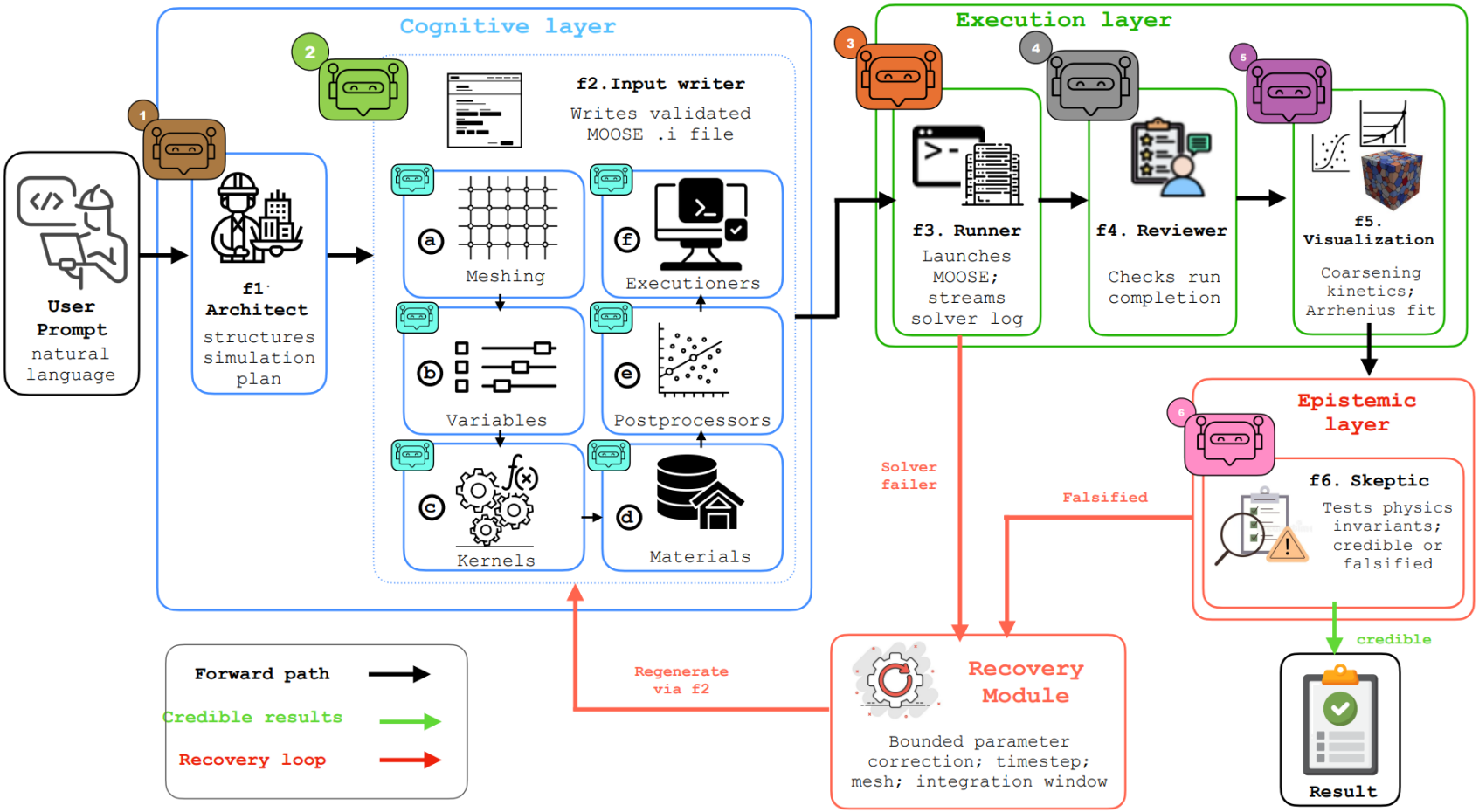


Figure 2: AutoMOOSE agentic pipeline.

Agentic Workflow

- The pipeline is a sequential composition of agent functions,

$$\mathcal{S} = f_6 \circ f_5 \circ f_4 \circ f_3 \circ f_2 \circ f_1(\mathcal{U}), \tag{8}$$

where $\mathcal{U}$ denotes the user input and each agent $f_i : \mathcal{P}_{i-1} \to \mathcal{P}_i$ progressively enriches a shared simulation plan,

$$\mathcal{P} = (\Omega, h, \mathcal{M}, \mathcal{B}, \boldsymbol{\theta}_{\text{solver}}, \boldsymbol{\theta}_{\text{run}}, \mathcal{O}), \tag{9}$$

which is the single shared state object of the pipeline.

- Workflow:
  1. Architect f_1: parses $\mathcal{U}$, produces a fully resolved plan $\mathcal{P}$.
  2. Input Writer f_2: reads $\mathcal{P}$, generates validated MOOSE `.i`.
  3. Runner f_3: uses $\boldsymbol{\theta}_{\text{run}}$ and validated `.i` to run simulations and record run directories/provenance.
  4. Reviewer f_4: uses $\boldsymbol{\theta}_{\text{solver}}$ and MOOSE log to screen for completion.
  5. Visualization f_5: reads CSV outputs in $\mathcal{O}$; extracts $N(t)$, fits $\tilde{k}(T)$, obtains Q_{fit}.
  6. Skeptic f_6: uses $\mathcal{M}$ to select invariants, tests completed runs, returns a credible/falsified verdict.

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Architect Agent

- **Role**: receives user input $\mathcal{U}$, produces simulation plan $\mathcal{P}$ as structured JSON.
- **Five parsing tasks**:
  1. Identify physics formulation: `GBEvolution` or `LinearizedInterface`.
  2. Identify spatial dimension: 2D QUAD4 or 3D HEX8.
  3. Identify boundary conditions: periodic or Dirichlet.
  4. Detect sweep intent: multiple values for sweepable parameters such as T, `grain_num`, or Q, and extract sweep range Θ.
  5. Determine solver strategy: adaptive timestepping and AMR on/off.
- **Robustness**: strict JSON-only output enforced by the backend; unsupported physics returns an error and lists available plugins, preventing ambiguous or unsupported plans from propagating.

Input Writer Agent

- **Role**: converts the Architect's simulation plan $\mathcal{P}$ into a complete, validated MOOSE `.i` input file by coordinating six sequential sub-agents, each responsible for one MOOSE block group.
- **DAG Order**: topological order

$$[\text{Mesh}] \prec [\text{GlobalParams}] \prec [\text{Variables}] \prec [\text{Kernels}] \\ \prec [\text{Materials}] \prec [\text{Postprocessors}] \prec [\text{Executioner}] \tag{10}$$

- **Validation**: assembled `.i` file is checked for block completeness, parameter types, and cross-reference consistency before being passed to the Runner.

Sub-agent	MOOSE blocks	Key parameters set	Depends on
(a) Mesh	`[Mesh]`	$n_x/n_y/n_z$, $L_x/L_y/L_z$, `elem_type` (QUAD4 or HEX8), `parallel_type`; enforces $h \leq w_{\text{GB}}/4$	Ω, h, `dim` from $\mathcal{P}$
(b) Variables	`[GlobalParams]`, `[Variables]`, `[ICs]`, `[AuxVariables]`	`op_num`, η_i fields, Voronoi or Random IC; `bound_value` (LinearizedInterface)	n_x, n_y from *(a)*
(c) Kernels	`[Kernels]`, `[AuxKernels]`	`GrainGrowth` or `GrainGrowthLinearizedInterface` module [7]; `BndsCalcAux`, `FeatureFloodCountAux`	`op_num`, η_i from *(b)*
(d) Materials	`[Materials]`, `[UserObjects]`	L, μ, κ via Eq. (4) [5] (`GBEvolution`) or parsed $L/\kappa/\mu$ (`LinearizedInterface`); `PolycrystalVoronoi`, `GrainTracker`	$\{\sigma, w_{\text{GB}}, M_0, Q\}$ from $\mathcal{M}$; mesh from *(a)*
(e) Postprocessors	`[Postprocessors]`	`TimestepSize`, `NumDOFs`, `NumElements`	Variables from *(b)*, UserObjects from *(d)*
(f) Executioner	`[Executioner]`, `[BCs]`, `[Adaptivity]`, `[Outputs]`	`PJFNK` [25], `bdf2`, `IterationAdaptiveDT`, ϵ_{nl}, ϵ_{l}, periodic BCs, AMR, CSV	$\boldsymbol{\theta}_{\text{solver}}$, $\boldsymbol{\theta}_{\text{run}}$ from $\mathcal{P}$

Figure 3: Input Writer sub-agents and MOOSE block responsibilities.

Runner Agent

- **Role**: simulation execution, output organization, and provenance recording.
- **Execution and Parallelism**: launches MOOSE as a managed subprocess. Expands sweep requests into n independent instances and dispatches them as concurrent threads.
- **Run Directory and Provenance**: creates a timestamped, self-contained run directory (copying the `.i` file) and records full provenance.
- **Live Monitoring and Exit routing**: streams solver stdout in real time; on exit, routes control by exit code: non-zero → Reviewer (failure path); zero → Visualization agent (success path).

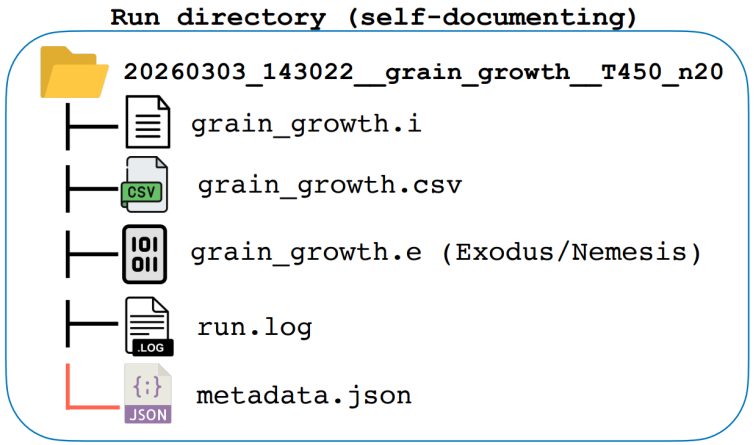


Figure 4: AutoMOOSE run directory structure.

Reviewer Agent

- **Role**: determines if a run executed successfully, distinguishing genuine solver breakdown from transient adaptive-timestep events.
- **Execution Screening**: checks tolerances in $\boldsymbol{\theta}_{\text{solver}}$: nonlinear residual $\|\mathbf{R}^{(k)}\|_2 < \epsilon_{\text{nl}}$ and linear Krylov residual $\|\mathbf{r}^{(k)}\|_2 < \epsilon_l$. Completion is read from the MOOSE log ("Finished Executing"), not just the exit code, avoiding misreading transient `DIVERGED_ITS` warnings as fatal.
- **Closed-loop Recovery** (`recovery.py`): bounded corrections. Primary: timestep cutback $\Delta t_0^{(k+1)} = \alpha \Delta t_0^{(k)}$, $\alpha = 0.5$. Other diagnoses map to bounded edits (e.g., extend window, refine mesh). Capped, logged, max 3 attempts per task.

Visualization Agent

- **Role**: extracts observables, fits kinetics, and returns numerical results with natural-language interpretation.
- **Kinetics Fitting**: extracts $N(t)$ from `grain_growth.csv`. Linearly regresses $N^{-1}(t)$ vs. t to obtain $\tilde{k}(T)$ for each $T_i \in \Theta$.
- **Arrhenius Regression and Consistency Check**: fits equation (7) to recover Q_{fit} and $\tilde{k}_0$. Q_{fit} must match user-specified Q in $\mathcal{M}$ (closed-loop check).
- **Narrated Output**: generates a structured natural-language interpretation.

Skeptic Agent

- **Role**: verification stage. Catches completed simulations that are physically wrong.
- **Method**: physics-grounded falsification battery. Failure = falsified; all-pass = credible. Each test paired with a diagnosis.
- **Invariants** (physics-specific): grain growth (non-conserved Allen–Cahn): T_1–T_5 (monotonicity, asymptotic consistency, parabolic scaling (Eq.(6)), Arrhenius consistency (Eq.(7)), numerical integrity).
- **Verdict & Recovery**: credible only if all invariants pass; otherwise falsified and routed to recovery module with diagnosis.

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Plugin Contract and Grain Growth Plugin

- Six-agent pipeline is **physics-agnostic**; connection to MOOSE physics via a minimal **plugin contract**.
- **Plugin Contract**: exactly two functions.
  - `generate_input(**params) -> str`: returns a complete MOOSE `.i` file; called by Input Writer.
  - `parse_results(csv_data: dict) -> dict`: returns observables such as $N(t)$, $\tilde{k}$, R^2, peak DOF, wall-clock statistics; called by Visualization.
- **Grain Growth Plugin**: first fully implemented plugin in AutoMOOSE. Supports two Allen–Cahn formulations, two IC types, 2D/3D geometries, and seven presets.

Plugin Extensibility and Spinodal Decomposition

- **Plugin Extensibility**: adding new physics requires only two functions, a `PLUGIN` metadata dictionary, and a physics-specific system prompt. No changes needed to orchestration, MCP server, or UI.
- **Spinodal Decomposition Plugin**: beyond grain growth, a second plugin for Spinodal Decomposition (conserved Cahn–Hilliard) is fully implemented and validated end-to-end on Fe–Cr alloy. Ferroelectric Switching and Solidification remain as registered stubs.

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Performance on Benchmarks

Agent	Stage	1st-attempt	Key metric
f_1 Architect	Intent parsing	1/1	Correct sweep intent, formulation, BCs
f_2 Input Writer	`.i` file generation	4/4	6/12 blocks exact match (Section 3.4)
f_3 Runner	Parallel execution	4/4	$T_{\text{sweep}} \approx 22{,}820$ s (1.8× vs. serial, scheduling only)
f_4 Reviewer	Completion screen	—	3/3 failed runs flagged (Section 3.7)
f_5 Visualization	Kinetics extraction	4/4	$R^2 = 0.90$–0.95 at $T \geq 600$ K (Section 3.5)
End-to-end		**4/4**	Prompt → kinetics analysis: ≈ 380 min

Figure 5: AutoMOOSE agent pipeline performance on the four-temperature grain growth benchmark ($T \in \{300, 450, 600, 750\}$ K). First-attempt success: correct output without Reviewer intervention. Timing on a MacBook Pro (Apple M-series, single core).

Performance and Skeptic Validation

Regime	n	G1	G2	G3	G4	G5	Valid & runs	Full
Core (T, N sweep)	8	8	8	8	6	8	8/8	6/8
Resolution / domain	4	4	4	4	4	4	4/4	4/4
Formulation (Lin. Interface)	4	4	0	4	3	4	0/4	0/4
Robustness (seeds)	4	4	4	4	2	4	4/4	2/4
3D	2	2	0	0	0	1	0/2	0/2
High-T stress	2	2	2	2	2	2	2/2	2/2
Density stress	1	1	1	1	1	1	1/1	1/1
Total	**25**	**25**	**19**	**23**	**18**	**24**	**19/25**	**15/25**

Figure 6: AutoMOOSE performance across the $n = 25$ benchmark regimes. G1: valid input generated; G2: ran to completion within the wall-clock window; G3: coarsened ($N_f < N_0$); G4: parabolic kinetics ($R^2 \geq 0.90$); G5: positive rate constant. *Valid & runs* counts G1–G3; *Full* counts all five gates. **Independent Skeptic falsifcation corroborates exactly the set of 15 credible runs.**

Quantitative Agreement of Coarsening Rate Constants

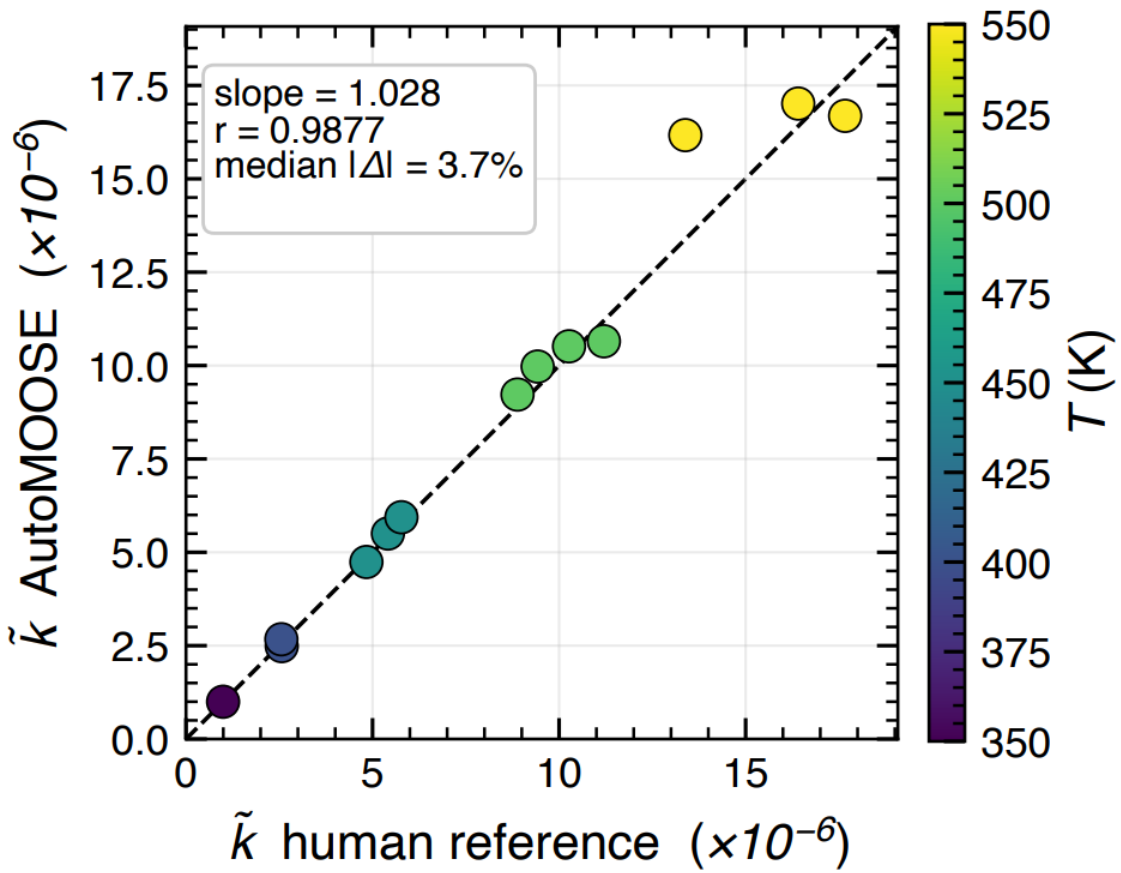


Figure 7: Matched-cell parity between AutoMOOSE-generated and human-reference runs. Across 14 matched grain-count/temperature cells, the agent-generated and human kinetic outputs agree closely.

Grain Shrinkage and Coarsening

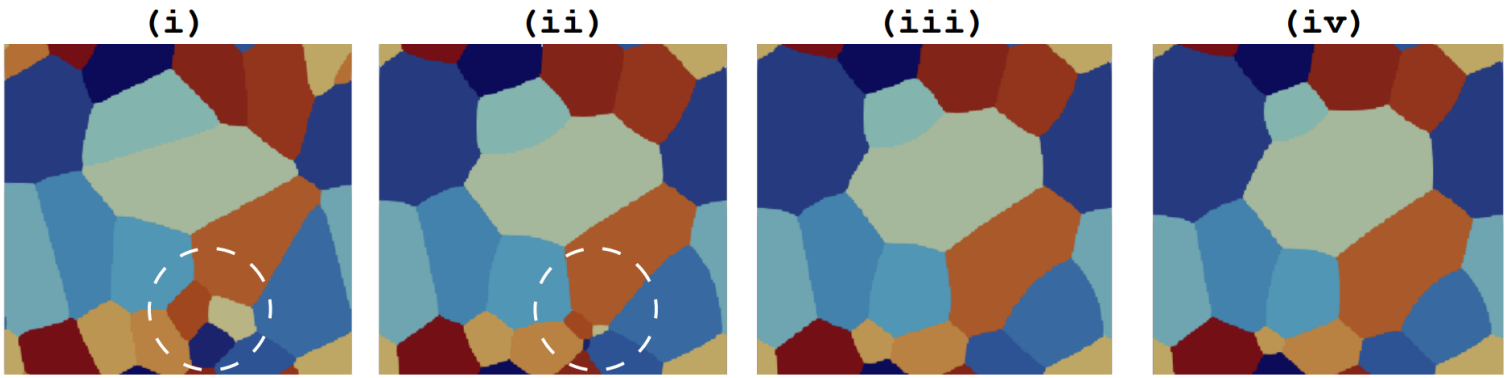


Figure 8: AutoMOOSE-generated microstructure evolution at $T = 450$ K. Panels (i)–(iv): phase-field grain structure at $t = 0, 500, 2000$, and 4000 ns. Grain count reduces from $N_0 = 15$ to $N_f = 13$ (Figure 9a). No user intervention was required.

Comparison Against Human Experts

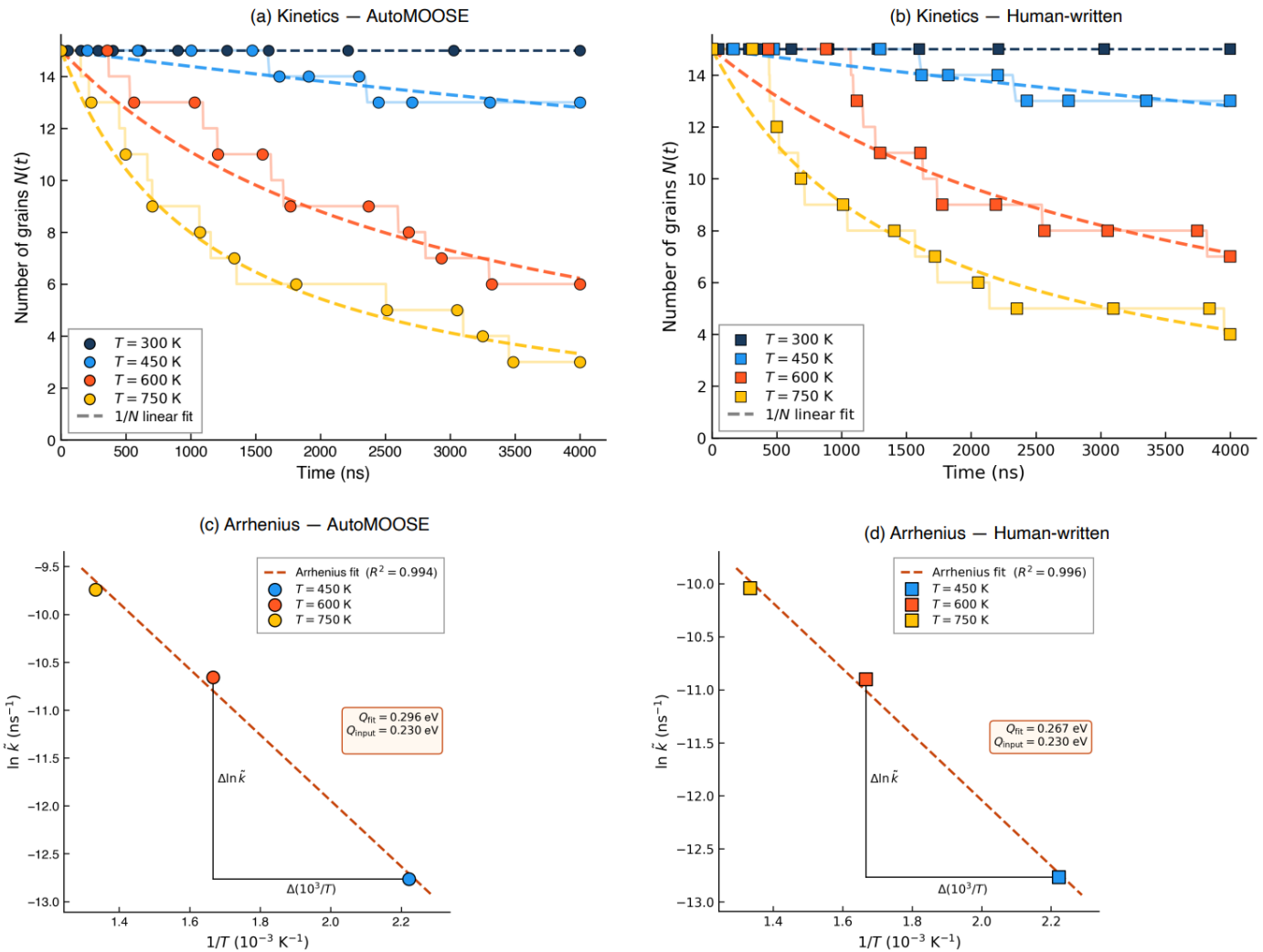


Figure 9: rain growth kinetics and Arrhenius analysis: AutoMOOSE vs. human-written reference.

Limitations

- Relies on pre-trained knowledge without RAG, making it susceptible to hallucinations for uncommon or recently added MOOSE modules.
- Recovery module handles only divergence-class failures via time-step cutback, unable to resolve complex root-cause formulation or numerical errors.
- Limited physics coverage (ferroelectric and solidification remain stubs), and results are subject to finite-size effects requiring large ensembles for accurate parameter recovery.

Section 2

Modular Large Language Model Agents for Multi-Task Computational Materials Science

Basic Information

- **Title**: *Modular Large Language Model Agents for Multi-Task Computational Materials Science.*
- **Code**: `https://github.com/cakshat/MatSci-LLM-Agents`.
- **Publication time**: Published online: 26 March 2026 (Received: 5 May 2025; Accepted: 14 October 2025).
- **Journal/Venue**: *Communications Materials* (Nature Portfolio), Vol. 7, Article 131 (2026); DOI: `https://doi.org/10.1038/s43246-025-00994-x`.
- **Authors**: Akshat Chaudhari[1], Janghoon Ock[1,2,*], Amir Barati Farimani[1,*]
- **Affiliations**: [1]Carnegie Mellon University; [2]University of Nebraska–Lincoln
- **Corresponding**: `jock2@unl.edu`, `barati@cmu.edu`

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Motivations

- Materials science data and simulation tools remain fragmented, limiting seamless integration and slowing materials design.
- General-purpose LLMs hallucinate, lack domain-specific knowledge, and cannot perform rigorous physics-based simulations alone.
- Existing LLM-agent frameworks are narrow in scope, creating a need for a general, modular, and extensible multi-agent system.

Prior Work: LLaMP

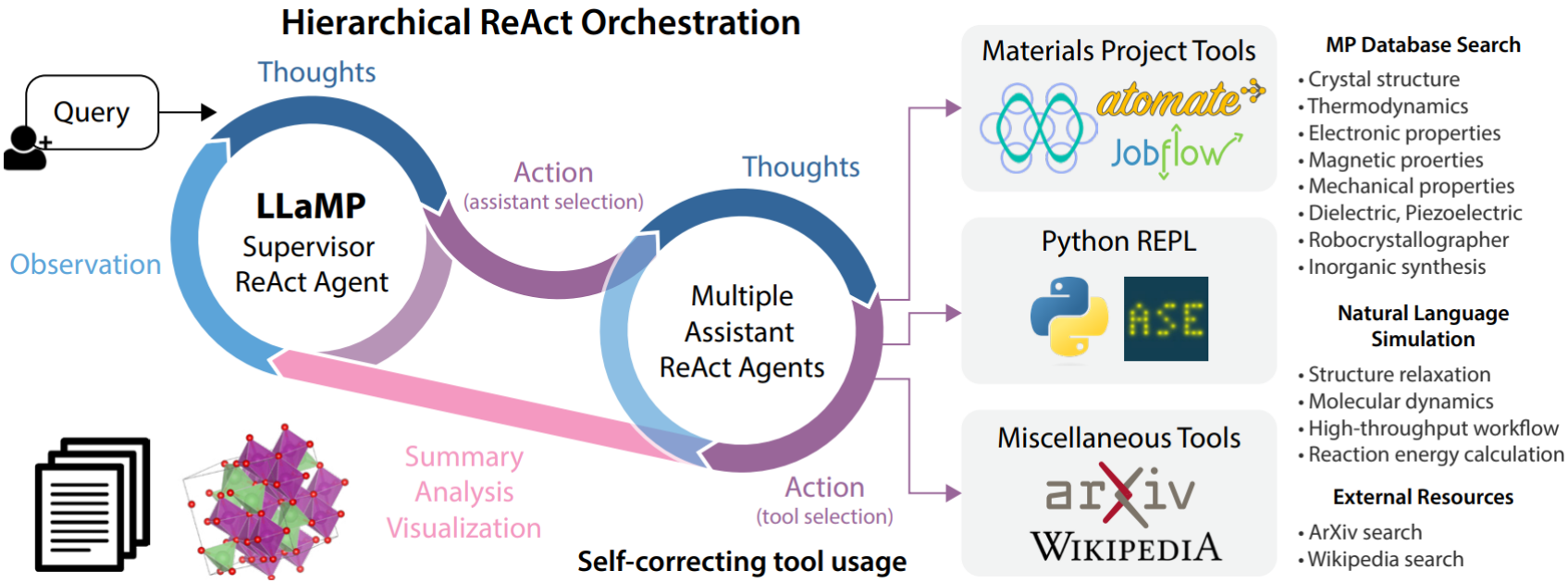


Figure 10: Hierarchical ReAct agent planning in LLaMP [Chiang et al., 2024].

Prior Work: HoneyComb

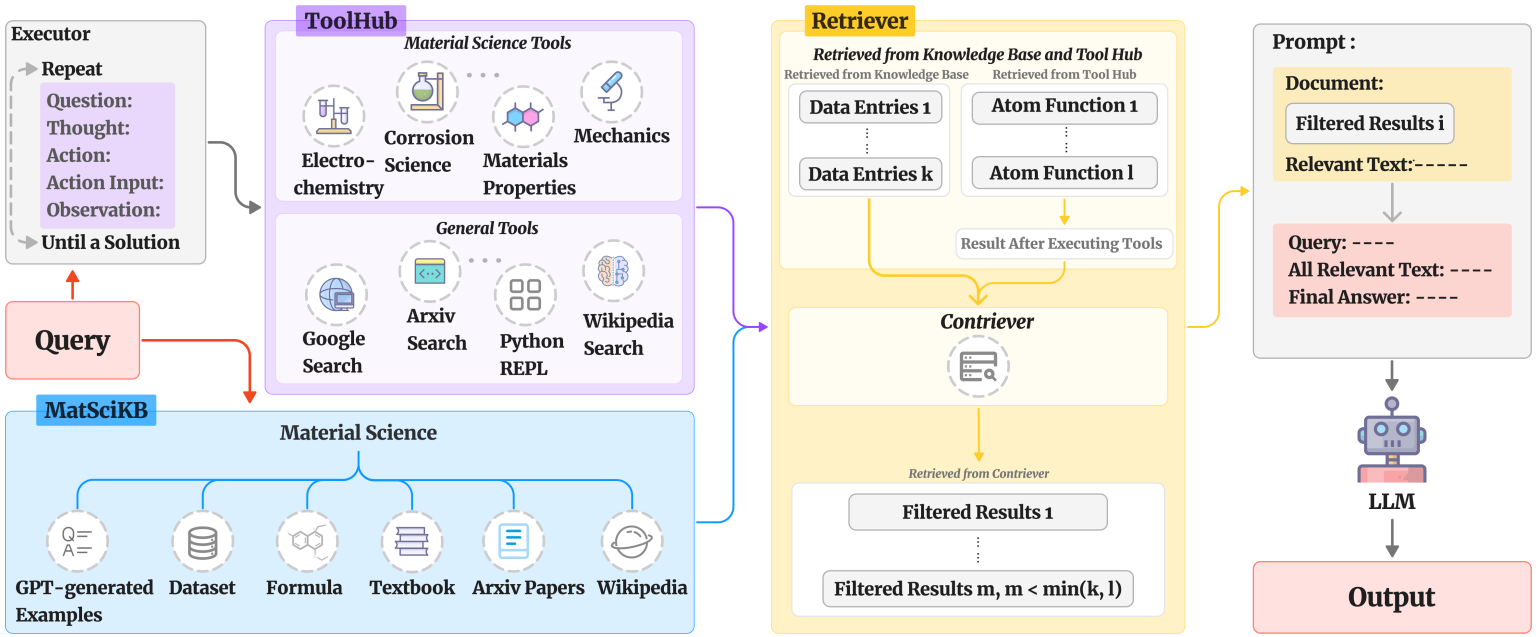


Figure 11: Framework of HoneyComb [Zhang et al., 2024].

Limitations

- Lack of continuum simulation and dedicated crystal generation capabilities.
- No task-specific multi-agent routing or cross-agent joint workflows for complex multi-step tasks.
- Insufficient reliability analysis: no repeated-run consistency checks for tool calls and parameter extraction.

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MatSciAgent: Multi-Agent System Architecture

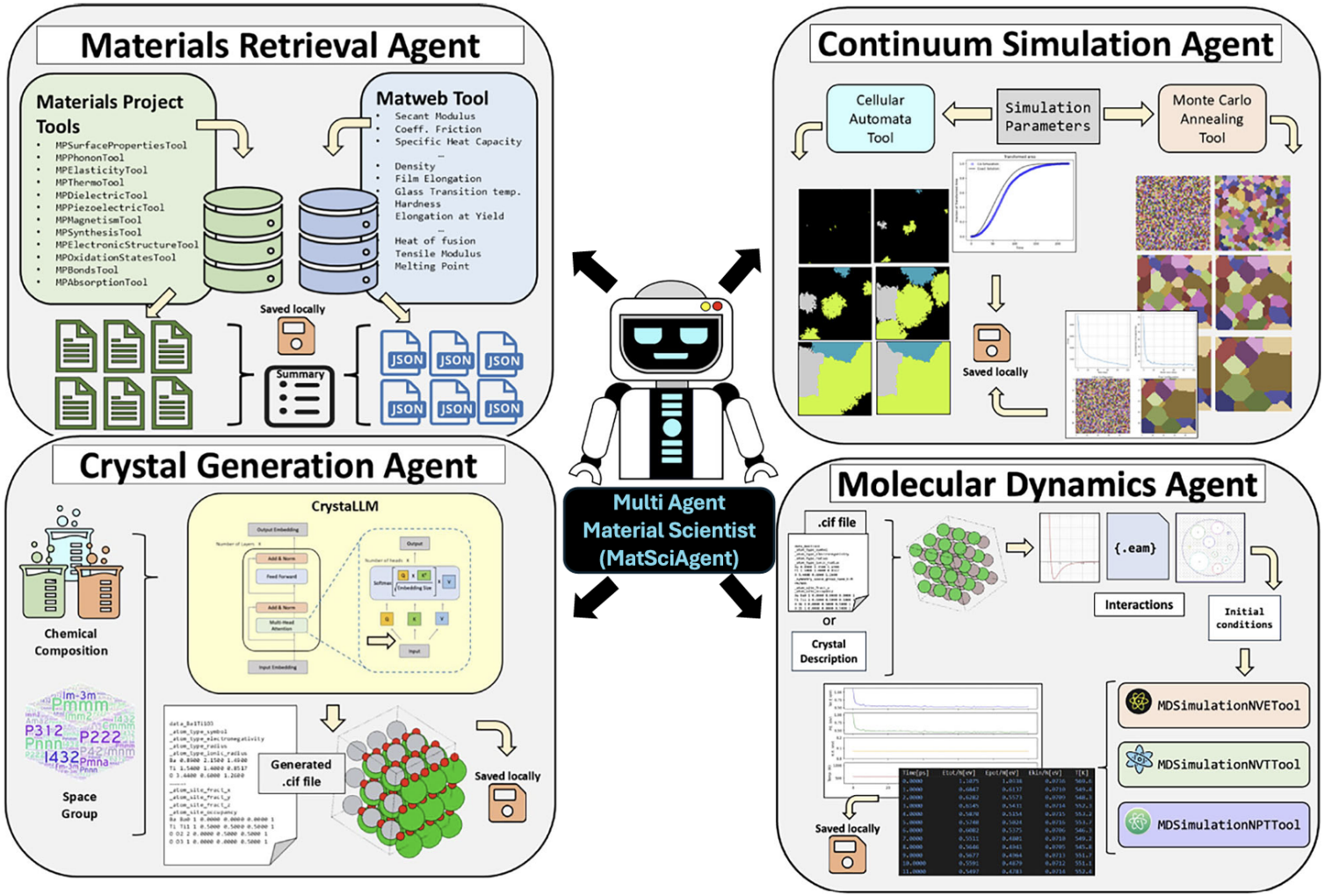


Figure 12: Framework of MatSciAgent. GPT-3.5-turbo for all agents (modular).

Iterative Execution Workflow of a Single Agent

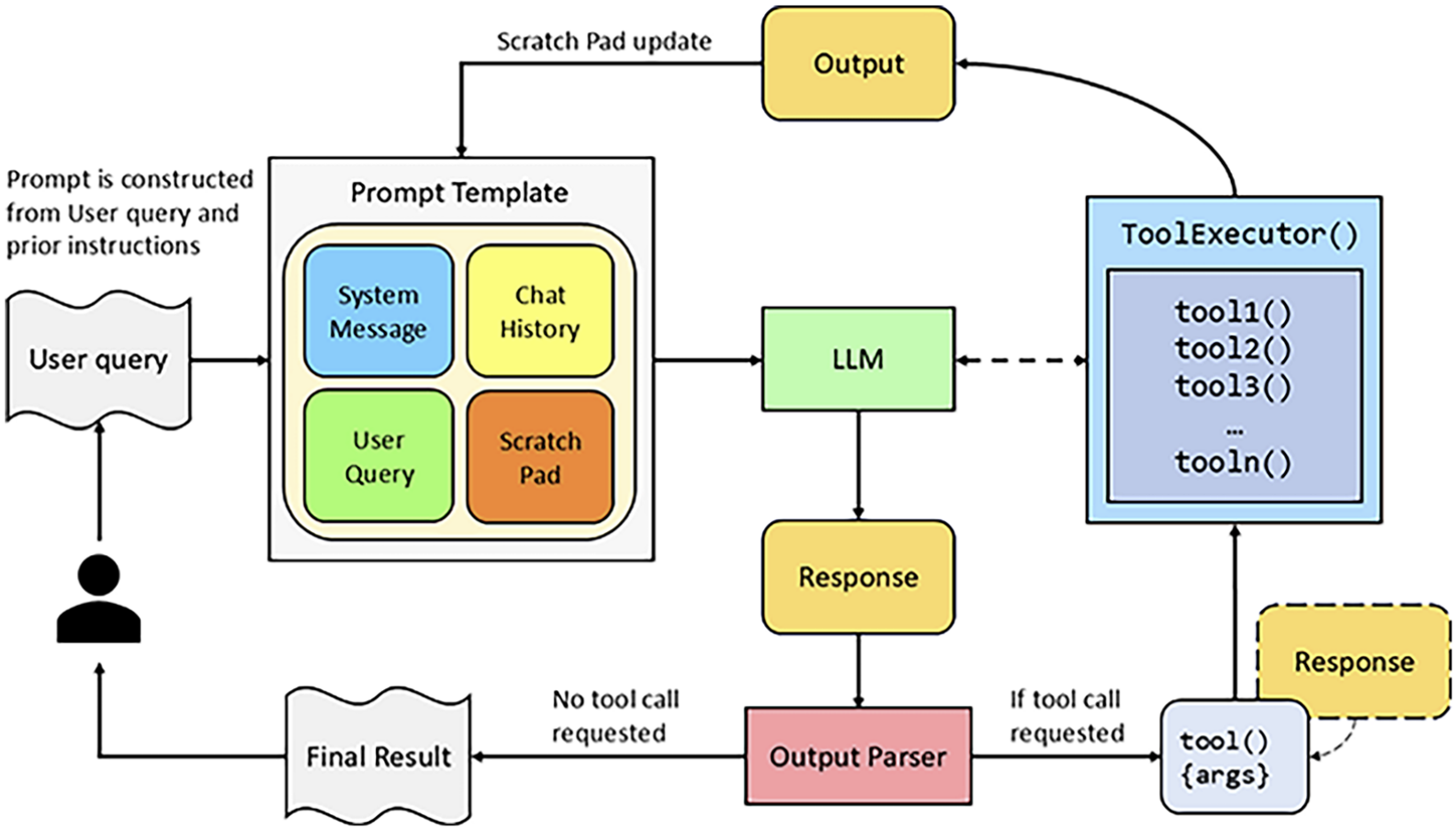


Figure 13: Components of an LLM agent.

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Materials Databases

- **The Materials Project**:
  - Open-access DFT-computed database: 200k+ inorganic materials, 577k+ molecules.
  - Properties: electronic, thermodynamic, elastic, magnetic, dielectric.
  - Free API and web access.
  - Used for materials discovery and ML training.
- **MatWeb**:
  - Commercial database of experimental/manufacturer data: 200k+ datasheets.
  - Covers metals, polymers, ceramics, composites.
  - Properties: mechanical, thermal, electrical, physical.
  - Free basic access; Premium for CAD/FEA export ($99.95/yr).

Materials Extraction Agent

- **Role**: Retrieve and summarize materials data from authoritative databases.
- **Tools**:
  - Materials Project API.
  - Custom web scraping for MatWeb.
- **Workflow**:
  - Parse user query $\rightarrow$ identify material/property.
  - Invoke database tools $\rightarrow$ retrieve validated data.
  - Synthesize grounded natural-language response.
- **Addresses LLM Limitations**:
  - Reduces hallucination.
  - Overcomes static pre-trained knowledge.
  - Provides domain-specific, up-to-date data.

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Cellular Automata

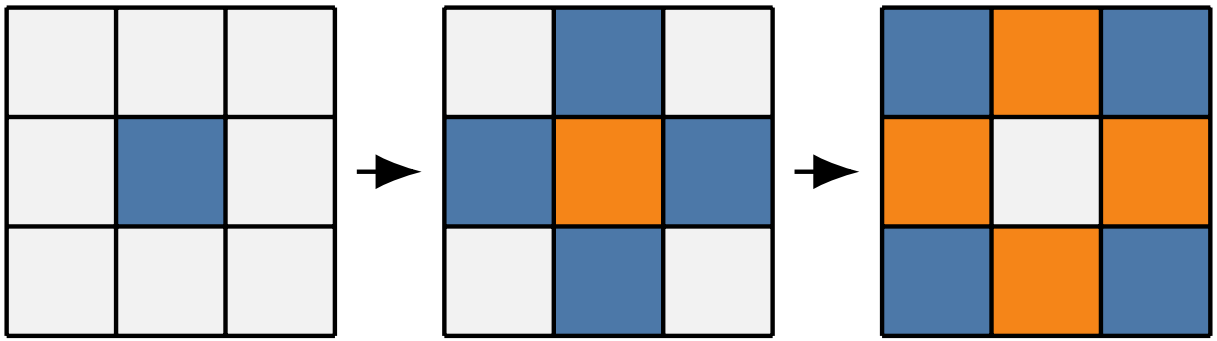


Figure 14: Evolution of a 3x3 cellular automata: A $\rightarrow$ B $\rightarrow$ C

- **Model**: 2D cellular automata for phase transformations (solidification/recrystallization). Cells are liquid (0) or solid (positive integers), driven by local interactions and attachment probability.
- **Parameters**: `width` (grid size), `attach_prob` (1=deterministic, <1=probabilistic), `num_solid_states` (1=solidification, >1=recrystallization).
- **Evolution**: `place_nuclei` randomly seeds initial solid cells. `evolve(Nsteps)` updates cells based on neighbor states.
- **Output**: microstructure snapshots and transformed area fraction over time, validated against JMAK theory.

Monte Carlo Annealing

- Simulates grain growth/spin ordering in a 2D Potts model at $T \approx 0$ using zero-temperature Metropolis dynamics, where spin flips are accepted only if they lower or maintain system energy.
- **Parameters**: Q states per lattice site (e.g., grain orientations), $L \times L$ lattice size, Monte Carlo steps (each with L^2 spin-flip attempts), and a biasing strategy for selecting sites and trial states.
- **Output**: energy history, acceptance history, and configuration snapshots for visualizing microstructure evolution; plots and saved images can be generated for further analysis.

Continnum Simulation Agent

- Executes advanced materials science simulations by integrating CA for phase transformations and MCA for microstructural evolution.
- Translates user-provided natural language simulation conditions into actionable parameters and automatically routes the task to the appropriate tool.
- Embedded within an LLM framework, it enables natural language interaction, parameter customization, and visualization, delivering predictive insights into microstructural stability.

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Crystal Generation Agent

- **Role**: Generate plausible crystal structures using CrystaLLM, a GPT-2 based LLM designed to produce CIFs of inorganic crystals [Antunes et al., 2024].
- **Workflow**:
  - Parse user query → extract composition and space group.
  - Invoke CrystalLLM → generate CIF file.
  - Return natural-language description and structure visualization.
- **Key features**:
  - Generates multiple candidate structures for a given composition.
  - Provides immediate input for downstream tasks (MD, stability analysis).
  - Modular: can accommodate other generative models, including property-constrained generation.

Recent Crystal Generation Models

Model	**Method**	**Year**	**Reference**
CrysLLMGen	Hybrid	2025	[Khastagir et al., 2026]
ADiT	Diffusion	2025	[Joshi et al., 2025]
CrystaLLM-π	Autoregressive	2025	[Bone et al., 2025]
CrystalICL	Autoregressive	2025	[Wang et al., 2025]
Materium	Autoregressive	2025	[Dobberstein and Hamaekers, 2026]
CrystalFormer	Autoregressive	2025	[Cao et al., 2025]
Matra-Genoa	Autoregressive	2025	[De Breuck et al., 2026]
deCIFer	Autoregressive	2024	[Johansen et al., 2025]

Table 1: Categorization of crystal generation models.

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ASE (Atomic Simulation Environment)

- Open-source Python toolkit for atomistic simulations.
- Supports MD in NVE, NVT, NPT ensembles; Berendsen thermostat/barostat for temperature/pressure control.
- Outputs: `.traj` trajectories, logs (energy, temperature, stress), and visualizations.

Molecular Dynamics Agent

- Automates molecular dynamics simulations from natural language queries.
- Built on ASE; supports NVE/NVT/NPT ensembles, EMT/EAM/LJ potentials, and Berendsen thermostat/barostat.
- Outputs: `.traj` trajectories, logs (energy, temperature, stress), and visualizations.

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Grounded Data Retrieval vs. LLM Hallucination

(a) User Query

Retrieve bond length and coordination information for TiO2. List your results.

(b) Materials Extraction Agent

Invoking: `get_material_ids` with `{'compound': 'TiO2'}` responded: This query falls under Type 2, which involves getting properties of a specific material. The specified material is TiO2, which corresponds to the chemical formula of Titanium Dioxide. To proceed, I will first need to obtain the material_id(s) for TiO2 by using the chemical formula. After that, I will extract the bond length and coordination information for TiO2. Let's start by retrieving the material_id(s) for TiO2. [MPID(mp-390), MPID(mp-430), MPID(mp-1439), MPID(mp-1840), MPID(mp-2657), MPID(mp-9173), MPID(mp-25262), MPID(mp-25433), MPID(mp-34688), MPID(mp-553432), MPID(mp-554278), MPID(mp-572822), MPID(mp-636827), MPID(mp-655656), MPID(mp-754672), MPID(mp-754769), MPID(mp-756587), MPID(mp-756655), MPID(mp-766454), MPID(mp-775938), MPID(mp-777129), MPID(mp-849533), MPID(mp-1008677), MPID(mp-1041565), MPID(mp-1044853), MPID(mp-1079207), MPID(mp-1101022), MPID(mp-1102255), MPID(mp-1102591), MPID(mp-1190753), MPID(mp-1201727), MPID(mp-1245098), MPID(mp-1245134), MPID(mp-1245308), MPID(mp-1245321), MPID(mp-1341203), MPID(mp-1350429), MPID(mp-1380335), MPID(mp-1386640), MPID(mp-1400963), MPID(mp-1408285), MPID(mp-1426806), MPID(mp-1445035), MPID(mp-2420244), MPID(mp-2739239), MPID(mp-2901430)]...

1. **Material Properties**:
   - **Crystal System**: Orthorhombic
   - **Volume**: Ranges from 113.66 $Å^3$ to 757.26 $Å^3$
   - **Density**: Ranges from 1.40 g/cm^3 to 4.56 g/cm^3
2. **Bonding Information**:
   - **Coordination Environments**: O2--Ti4+(3) and Ti4+-O2-(6)
   - **Bond Lengths**: Range from 1.88 Å to 2.20 Å
   - **Bond Types**: O2--Ti4+ interactions 3. **Crystal Structure**:
   - **Space Group**: Pbcn, Pbca, Fd-3m, P4_2/mnm, C2/c, I4_1/amd, Pnma, C2/m, P3_121, Fd-3m, Pmm2, Cmm2, Pnma
4. **Material IDs**:
   - mp-390, mp-430, mp-1439, mp-1840, mp-2657, mp-9173, mp-25262, mp-25433, mp-34688, mp-553432, mp-554278, mp-572822, mp-636827, mp-655656, mp-754672, mp-754769, mp-756587, mp-756655, mp-766454, mp-775938, mp-777129, mp-849533, mp-1008677, mp-1041565, mp-1044853, mp-1079207, mp-1101022, mp-1102255, mp-1102591, mp-1190753, mp-1201727, mp-1245098, mp-1245134, mp-1245308, mp-1245321, mp-1341203, mp-1350429, mp-1380335, mp-1386640, mp-1400963, mp-1408285, mp-1426806, mp-1445035, mp-2420244, mp-2739239, mp-2901430

(c) GPT-3.5-turbo

In titanium dioxide (TiO2), the bond length of the Ti-O bond is approximately 1.95 Å (angstroms).
In terms of coordination, Ti in TiO2 typically exhibits octahedral coordination, where each titanium atom is surrounded by six oxygen atoms arranged in an octahedral fashion.

Figure 15: Material-specific information retrieval. (a) User query specifying the target material. (b) Output from the materials extraction agent. (c) Output from GPT-3.5-turbo.

CA Simulation of Solidification

(a) User Query

Simulate the solidification of an Al-Cu-Mg alloy system using a CA model. The system should be modeled on a square domain with a width of 100 grid units to ensure adequate resolution of microstructural features and multi-phase interactions. Set the attachment probability to 0.2, representing realistic diffusion-controlled growth processes for metallic systems under typical cooling rates.

(b) Continuum Simulation Agent

> Entering new AgentExecutor chain...
Invoking: `CellularAutomataPhaseTransform` with `{'width': 100, 'attach_prob': 0.2, 'num_solid_states': 3, 'num_nuclei': 5}`The simulation for the solidification of an Al-Cu-Mg alloy system using a Cellular Automata model on a 100x100 grid with an attachment probability of 0.2 has been initiated. The system is set to have 3 solid states and 5 initial nuclei for the transformation process.

> Finished chain.

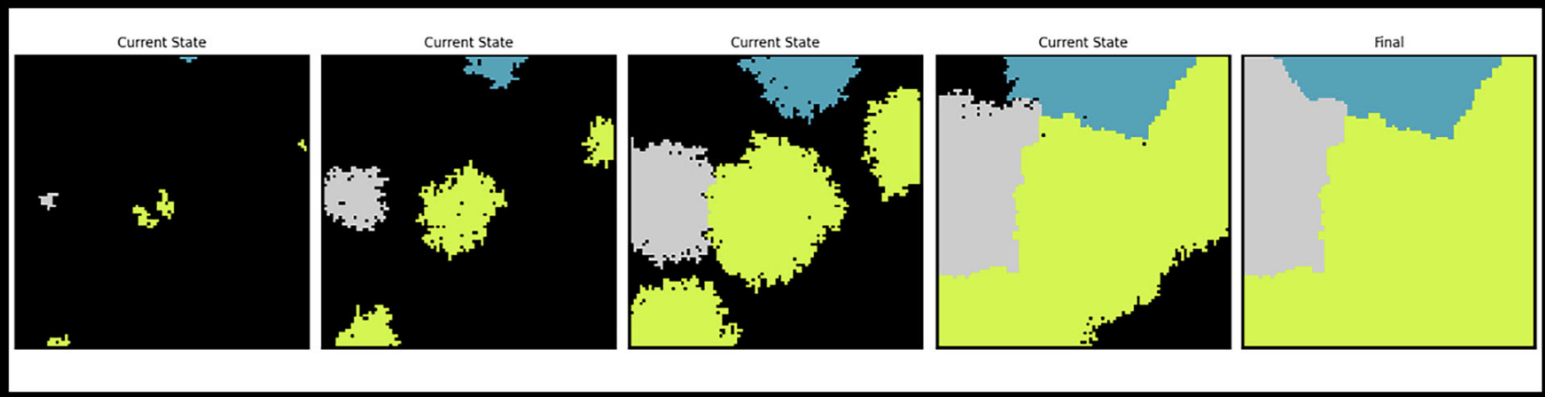


Figure 16: Cellular Automata simulation of diffusion-controlled growth. (a) User query specifying the simulation task. (b) Agent-generated output, including frame-by-frame microstructure evolution.

MCA Simulation of Grain Growth

(a) User Query

Using Monte Carlo annealing at T ~ 0K simulate grain growth in a polycrystalline aluminum sample, to model the minimization of grain boundary energy duringannealing. Use a 50x50 lattice grid, with each site representing a distinct grain orientation chosen from 20 possible flavors. Begin by initializing the lattice with random grain orientations and identifying grain boundaries. Focus updates on interface sites, where neighboring grains meet, to simulate energy reduction at grain boundaries. Perform 100 Monte Carlo steps, dynamically tracking and updating the positions of interface sites to reflect grain coarsening. Capture snapshots of the lattice at regular intervals to observe the evolution of grain structure, including the growth of larger grains at the expense of smaller ones.

(b) Continuum Simulation Agent

> Entering new AgentExecutor chain...

Invoking:` BiasedPottsMonteCarloAnnealing` with `{'Q': 20, 'L': 50, 'MC_steps ' : 100, 'biasing_strategy': 3}`

> Finished chain.

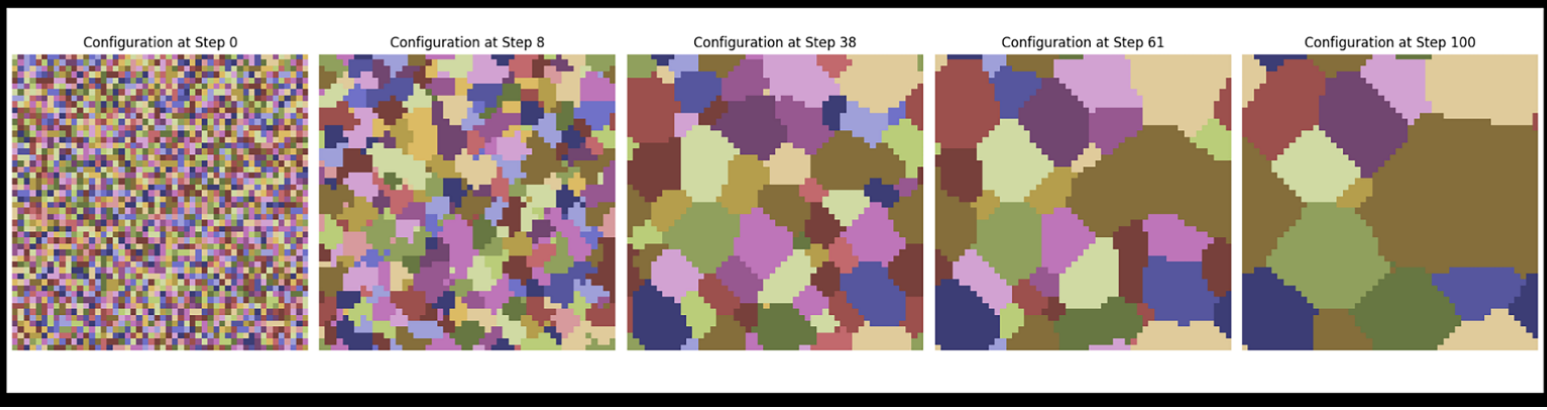


Figure 17: Monte Carlo Annealing simulation of polycrystalline grain growth. (a) User query specifying the simulation task. (b) Agent-generated output, including frame-by frame microstructure evolution.

Crystal Structure Generation

(a) User Query

Create a CIF file for a material with the chemical composition Sr2FeMoO6 in the double perovskite structure.

(b) Crystal Generation Agent

> Entering new AgentExecutor chain...
Invoking: `CrystaLLM` with `{'composition': 'Sr2Fe1Mo1O6'}`
The cif file provides data for the compound Sr2FeMoO6 with crystallographic information. The crystal structure belongs to space group P4/mmm with cell parameters a=b=5.65 Å, c=3.9733 Å, and angles alpha=beta=gamma=90°. The compound has a chemical formula Sr2Fe1Mo1O6, where Sr, Fe, Mo, and O occupy specific fractional coordinates in the unit cell. The atom types have different electronegativity and ionic radius values. This information can be used for further analysis and simulation of the compound's properties. However, there is a warning that the file was not post-processed properly by CrystaLLM, indicating potential issues with the data that need to be addressed before using it for in-depth investigations. The CIF file for the material with the chemical composition Sr2FeMoO6 in the double perovskite structure has been generated.

> Finished chain.

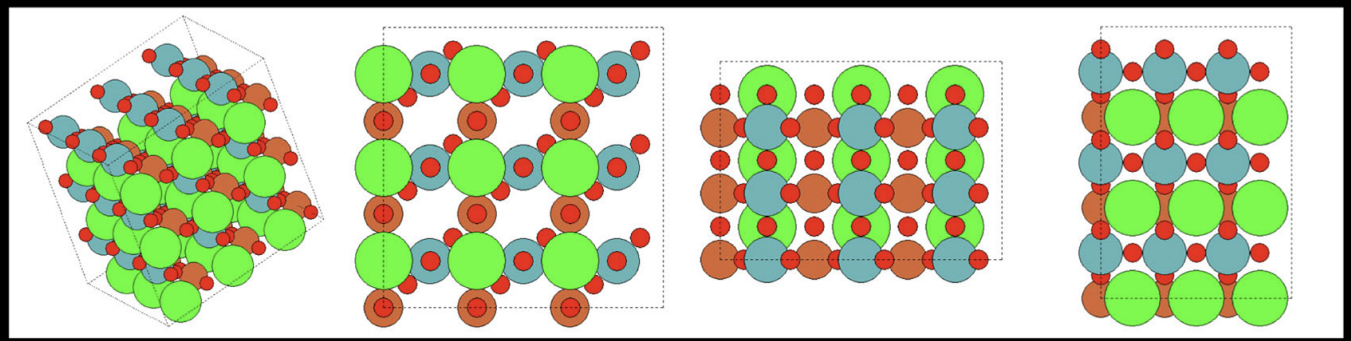


Figure 18: Crystallographic generation for Sr_2FeMoO_6. (a) User query specifying the chemical composition and application. (b) Agent-generated CIF file description and corresponding visualization of the crystal structure.

Molecular Dynamics Simulation

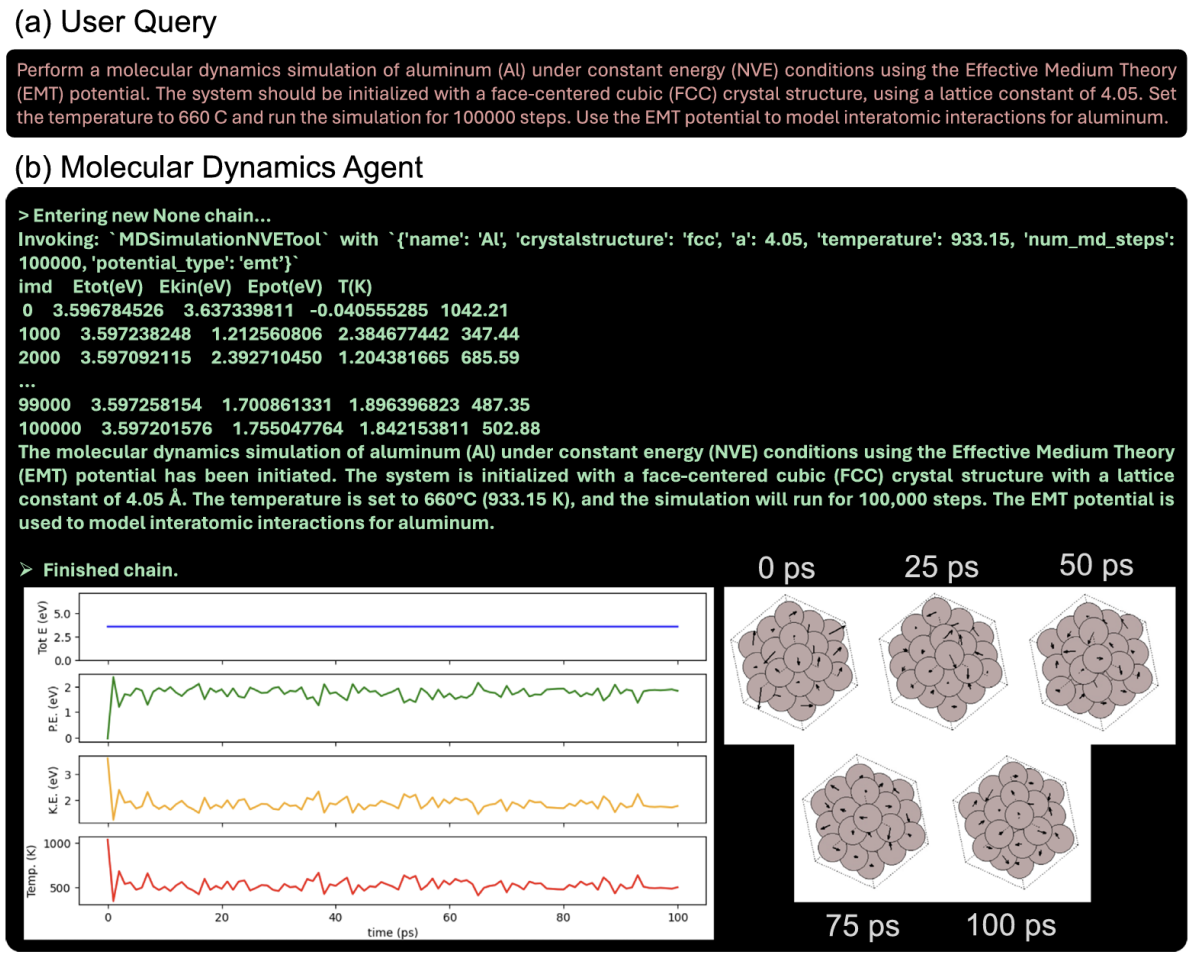


Figure 19: MD simulation of aluminum. (a) User query specifying the simulation conditions, including ensemble type and temperature. (b) Agent-generated output showing a summary of the simulation setup and the resulting MD trajectory.

Joint Task

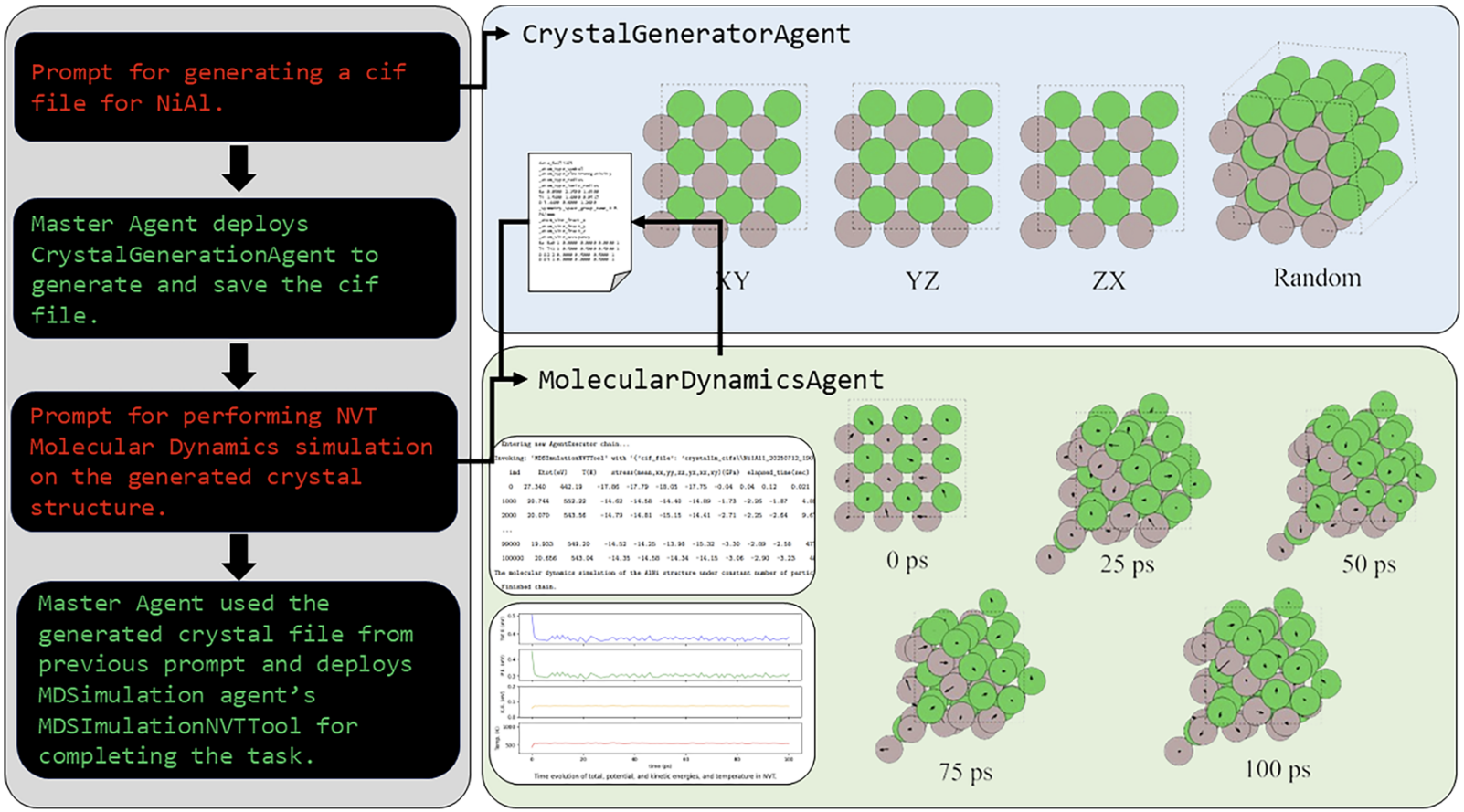


Figure 20: Coordinating multiple agents for joint task. After generating the CIF file the master agent successfully deployed MolecularDynamicsAgent to use the cif file generated previously to run the MD simulation.

Limitations

- **Restricted Simulation Scope**: continuum simulations are limited to 2D grids. MD simulations only support EMT, EAM, and LJ potentials, lacking dynamic code generation for unseen tasks.
- **Data and Generation Constraints**: relies solely on Materials Project and MatWeb. CrystalLLM outputs may contain post-processing warnings, requiring human validation before use.
- **LLM and Reliability Limits:** Relies on vanilla GPT-3.5-turbo without fine-tuning. Exhibits isolated extraction failures (e.g., Run 5) and only 80% consistency in output formatting.

Section 3

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Thank you!
Any questions?